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RESEARCH-ARTICLE

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Abstract

In this work, we designed and developed *Dice Adventure*, a turn-based multiplayer game where three characters work together to reach their individual goals and then a shared team goal to complete each level. Using *Dice Adventure* as the environment, we hosted a game competition with two tracks: agent and player. Participants could join one or both by submitting an agent they developed and/or signing up to play with the agents submitted by other developers. We collected competition game play data as part of a human-AI teaming pilot study to understand team behaviors, performance, and to test our systems. Insights from the competition also informed the design of a randomized controlled study for future experiments and competitions, aimed at exploring key human-AI teaming questions—such as how role assignments, team compositions, and team dynamics influence team performance. Our work introduces a novel gaming environment to support future research on human-AI teaming and offers preliminary insights into the design of such studies.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **User studies**.

Keywords

Human-AI teaming, multiplayer game, team dynamics, study design

ACM Reference Format:

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1 Introduction

For several decades, collaborations between humans and machines have been the driving force of progress in various domains including

healthcare, education, military operations, and video games. Recent advances in Artificial Intelligence (AI) have enabled machines to achieve superhuman abilities in various tasks, including game play [19, 28]. However, AI systems still fail in complex problems where human-AI collaborations are unavoidable and necessary. Past studies show that AI agents still generally lack the ability to recognize the knowledge and behaviors of their teammates [32], as well as to adapt their responses to previously unseen situations [10]. For example, the OpenAI five agents [2] were impressive when playing DOTA2 as an AI team against a human team, but struggled to adapt their strategies when playing on mixed human-AI teams.

To advance research on AI adaptability and raise awareness of human-AI teaming, we hosted a competition using *Dice Adventure*,¹ a turn-based, multiplayer game we developed where three players take on the roles of characters with different capabilities. In this game, players must work together to traverse maze-like dungeons by achieving both individual and team goals to clear each level. Our competition had two tracks, an agent track and a player track. In the agent track, participants submitted AI agents to play the three *Dice Adventure* characters. People could also playing the game without submitting an agent. We developed a matchmaking system to automatically match human players with AI agents as well as randomly assign roles to human players and AI agents. We also offered a low-fidelity Python version of the game for participants to develop and test their AI agents. Lastly, we provided two baseline agents, a pre-trained Reinforcement Learning (RL) agent and a Hierarchical Task Network (HTN) agent for people to play with.

As part of our competition, we also ran a pilot study to test our systems and investigate human-AI teaming within this game. As part of our study, we collected game data as well as game play observation field notes collected at the in-person competition sessions. We also collected team performance data, including the total score they obtained at the end of a game, after a team cleared all levels or lost all lives. The three best teams were awarded cash prizes provided by IEEE Computational Intelligence Society Education Competition Fund.

In this paper, we start by introducing the *Dice Adventure* game with details on the systems we developed and their capabilities. Afterwards, we present the competition design with subsections describing the rules, evaluation criteria, and proposed experimental designs. We then present the competition results with preliminary

¹<https://github.com/STRONG-TACT/HMT-Game-1>



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analyses and observations on game strategies employed by the participants. In future iterations of the competition and experiment, we want to understand how communication, coordination and adaptation affect a team's performance. We conclude with a discussion of contributions, limitations, and future research directions.

Our work aims to investigate the following questions:

- How can cooperative multiplayer games and game competitions be leveraged to investigate human-AI teaming?
- How does role assignment and team composition affect overall team performance?
- How does role assignment and team composition affect the strategies and behaviors that a team employs?

The *Dice Adventure* Competition was also an opportunity to bring students, researchers, and developers together to explore human-AI teaming futures. Moreover, the game itself could serve as a testbed for future researchers to run studies on teaming. It could also benefit CS education by serving as an environment for students in Artificial Intelligence (AI), Knowledge-Based AI, Game AI and Deep Learning classes to learn about agent design and development. We plan to run future iterations of the *Dice Adventure* Competition with more types of agents to better understand communication, collaboration and adaptation in human-AI teams.

2 Background

2.1 Human Machine Teaming (HMT)

Early research on human-machine teaming emphasized the goal of improving overall team performance [3, 6], with psychologists and engineers exploring how machines could augment human performance [6, 7] and automate redundant and repetitive tasks [1]. The discussion of human-machine teaming gradually shifted from “machines as tools” to “machines as teammates” [20] since machines are becoming socially important as technology advances [12].

Previous studies have shown that effective human-machine teaming is built upon the success of complex and bidirectional interactions between human and machine agents, and between these agents and their environment [14, 18, 27]. However, people may perpetuate misconceptions and oversimplification of AI's roles in human society by comparing human/AI capabilities and limitations arbitrarily, which limit the potential use cases of AI [16]. To bridge this gap and to raise the awareness of human-AI teaming, we propose the use of *Dice Adventure* as a testbed for human-AI teaming experimentation. Our aim is to explore how role assignments, team compositions and team dynamics affect team performance.

2.2 Multiplayer Games to Study HMT

Multiple studies have utilized the popular multiplayer game *Overcooked!* and *Overcooked!2* to study human-AI teaming behaviors. Rosero and team [23] discussed the effectiveness of communication in human-human teamwork and human-agent teamwork by running a Wizard of Oz experiment. Long and team [15] conducted a proof of concept study on how agents could adapt to humans based on a trust score measured via eye tracking.

However, we recognized that the *Overcooked* environment cannot help us to understand how communication, coordination, and adaptation are established in human-AI teaming scenarios. For one

reason, all objects are observable in this environment, with teammates sharing vision, goals, and capabilities. In this case, teamwork can increase the team's efficiency but a single player can also complete all the tasks. We want to understand how different types of teamwork can be established in an asymmetrical environment where each team member has unique abilities. More importantly, we are interested in exploring bidirectional behaviors between humans and agents. For example, during the teamwork, is it possible for humans or agents to observe each other's behaviors and make adaptations? How can we evaluate the synchronization in teamwork when there are different communication protocols? Such observation can hardly be made in *Overcooked* as the steps to reach a goal can be well decomposed. With these motivations, we developed *Dice Adventure* so we could study more complex and unpredictable human-agent teaming behaviors.

2.3 Game AI Competitions

In addition to their entertainment value, games have long been used as popular benchmarks; e.g., as testbeds for AI systems [11] and as platforms for CS education [5, 13]. There are multiple game AI competitions [19, 21, 25, 29] hosted at prestigious conferences providing environments for participants to deploy their models, run against benchmarks and compete for prizes. However, few competitions have focused on human-AI teaming or provided environments in which humans and agents can collaborate—which is essential for studying teaming.

With the motivation of studying human-AI teaming effects, we employed speculative game design and developed *Dice Adventure*, which supports various teaming scenarios [8]. Beyond providing an interesting and dynamic environment for participants to submit and play with agents, we also wanted to pilot a human-subject study during our *Dice Adventure* Competition to better understand various human-AI teaming effects along the dimensions of communication, coordination and adaptation [27]. We aim to use insights discovered from the pilot to design a randomized controlled trial to further explore team dynamics, role allocation, and synchronization in human-AI teams [27, 31]. Our competition brought together students, professionals, and other enthusiasts of AI, Game AI, Deep Learning, Reinforcement Learning, Machine Learning and general Computer Science, by providing a space for them to develop and team with agents.

3 The Game: *Dice Adventure*

3.1 Multiplayer System

Dice Adventure implements reliable and scalable online multiplayer gameplay through the Photon PUN2 SDK, a well-trusted and established multiplayer server solution for our game engine of choice, Unity. Our current game server supports up to 100 concurrent players, or up to 33 concurrent game sessions running simultaneously. By facilitating multiple independent sessions, researchers can gather data and insights from diverse and geographically dispersed user groups, reducing the time typically needed to conduct numerous experiments for reliable human-AI teaming conclusions.

To support the multiplayer experience, we implemented a match-making system with a waiting room. The system can randomly

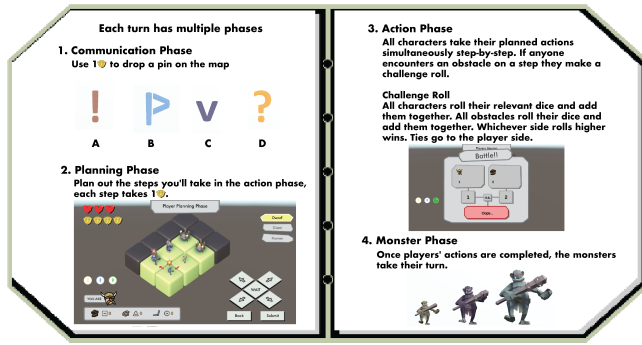


Figure 1: The four phases in a round. A level of the game may take multiple rounds to clear.

create teams for human players and AI agents with random role assignments. The AI agents on a hybrid team were randomly selected from a pool of available agents. The matchmaking system can be flexible in handling specific needs to cater to different research goals. For example, researchers can apply different matchmaking policies such as fixing the roles for human/AI players or controlling the number of AI agents on a hybrid team. Such configurations can be updated and synchronized to all players simultaneously.

3.2 Logging System

In HMT studies, researchers often need to examine and replay a single session for more comprehensive observations, or to debug AI agents to better understand how they are collaborating with and adapting to their human/AI teammates. To support this need, *Dice Adventure* implements a comprehensive game state and game action logging system. Every state change or player action is associated with a unique event ID. When an event is triggered, the corresponding ID will be sent to the configuration server along with a time stamp and all other relevant information (such as game state, logging message, etc.). This system enables researchers to reconstruct past events to replay the game in every session, and perform detailed post-analysis of teaming and decision-making.

3.3 Game Rules

Dice Adventure is a turn-based, multiplayer adventure game developed at Carnegie Mellon University. Three players are assigned to characters—*dwarf*, *giant*, and *human*—to collaborate and conquer a series of maze-like dungeons. The game has four *phases*: the communication phase, planning phase, action phase, and enemy (monster) phase. All four phases comprise a *round* and the players will take multiple rounds to complete each level as a team. Figure 1 shows the four phases and what players are expecting in each phase. Players can use *action points* to place pins on the map during the communication phase. After all the players submit the pins, they will plan the paths they want to take during the planning phase. When all players finish planning steps, they will move together during the action phase. If characters run into any challenge (rock, trap or monster), a combat will be triggered and they will roll their unique dice (shown in Figure 3) to fight against the challenges. The character will win the combat if their dice numbers are greater or



Figure 2: The components in the first level from *Dice Adventure*'s practice mode. Each level has a different map with shrines, tower, challenges and terrains. Later levels may take more time and rounds for the team to clear with more complicated designs.

equal than the dice number shown on the challenge's side. If two characters run into the same challenge, the dice numbers of the two characters will be added.

For each level of the dungeon, the players must navigate through the unknown terrain and find the *individual goal* (represented as a shrine in different colors) assigned to their character. After all players have reached their individual goals, one of them must reach the final goal (represented as a tower) to complete the level. Players may encounter obstacles and enemies as they explore the map. Figure 2 shows the objects in level 1 as an example. Below are the explanations to each annotation in the game scene.

- a: Health Points, 3 by default but can be configured to differ by character, health is reset at the beginning of each level
- b: Action Points, differs by character
- c: Button to forfeit a run
- d: Indicates which character the player is controlling
- e: A banner showing the status of all three characters
- f: A dialogue bubble indicating whether a player is still acting in the phase or is ready to move on
- g: Lives; each character has a set number of lives for the game, if any character runs out of lives the game will end at the current level
- h: An indicator showing whether a player has achieved their personal goal
- i: The current phase in the round
- j: The current round on this level, teams score more points if they complete levels in fewer rounds
- k: Phase countdown timer, if a player does not complete all of their actions in time, then the game will automatically submit for them.
- l: The characters—*dwarf*, *giant*, and *human* (left to right)
- m: Button to commit the character to end the current phase
- n: A personal goal (shrine), the color of each shrine corresponds to the character that must reach it to advance
- o: The final team goal (tower)
- p: A small monster
- q: A medium monster

There are 3 different characters				
	Dwarf	Human	Giant	
Health	3	3	3	
Action Points	6	4	3	
Vision Range	3 x 3	5 x 5	7 x 7	
	+ 0	+ 0	+ 0	
	+ 0	+ 0	+ 0	
	+ 0	+ 0	+ 0	

Your Goal	
Activate Each Shrine	then Reach the Tower

Obstacles		
Rocks Block your path until broken	Traps Can be disarmed but might hurt if they go off	Monsters Move around the map and

Figure 3: A description of characters, goals, and challenges.

- r: Impassible terrain where the characters cannot place pings or move
- s: A set of controls to allow players to switch between characters, the content in the dashed square is only visible in practice mode of the game

The game encourages communication, coordination, and adaptation between players through a set of asymmetric designs. Figure 3 shows the capabilities of the three characters, their goals, and the types of challenges they encounter. A detailed description of the *Dice Adventure* rules, characters, and challenge types can be found in this master rulebook on the competition website.²

3.4 Game Design

Dice Adventure was designed using speculative design approach [8] to explore human-AI teaming behaviors with a particular emphasis on asymmetric cooperative game mechanics. We incorporated three aspects of asymmetry when designing *Dice Adventure* to make it more interesting, challenging, and cooperative.

Information Asymmetry: The game includes a fog-of-war system (shown in Figure 4) that restricts the vision range of each player limits their ability to keep track of the game state. Additionally, while players are not allowed to communicate through text or audio, the game offers a pinning system for team communication in the first phase of each round. Players can place pings on the map as indicators to other players. The pings are symbols with no predefined semantics. As a result, the challenge of this system is that players must learn how their teammates use and interpret the pings during the game.

Ability Asymmetry: During the game, the players will encounter challenges such as enemies and resolvable obstacles. Encountering a challenge triggers a skill check, where players must roll over a target value to succeed with the stats of each character determining the size of dice rolled (shown in Figure 3). Thus the probability of winning a battle depends on the unique capabilities of each character. For example, the *giant* rolls an 8-sided die (D8) when trying to remove a rock while they only roll a 4-sided die (D4) when trying to disarm a trap.

Role Asymmetry: Taking both information and ability asymmetry into account, *Dice Adventure* creates a role asymmetry environment that balances the game and introduces multiple potential



Figure 4: In the fog-of-war system, the lighter gray cells are the areas the character has seen but are currently out of vision, where static elements (e.g., terrain, shrines, etc.) is visible but challenges are not. The darker cells are the areas the character has not yet seen, where neither terrain nor challenge is visible.

teaming scenarios. For example, we offer the *giant* the greatest vision but restrict its number of action points each round. In contrast, the *dwarf* has excellent agility with more action points but is given a very limited view of the map. Such asymmetry might allow for role designations such as “field agents” and “over watcher”. We hypothesize that human-AI teams will perform differently with different role assignment, team compositions (number of AI agents on a team), and team dynamics.

4 Agent Development Interface

4.1 Training Environment

In order to provide an easy way for users to interface agents with our game, we developed a custom, OpenAI gymnasium environment that supports connection to the Unity version of the game. We also released a “low-fidelity” Python implementation of the game to support agent training. The custom environment provides a common interface to both the Unity and Python versions of the game, with additional benefits to users wishing to train model when developing their agents. First, the environment supports standard, off-the-shelf reinforcement learning (RL) models from the Stable Baselines 3 python package [22], providing implementations for the *step()* and *reset()* functions, among others. If users wish to interface with the Python version of the game, it has the added benefit of near-clock speed game play, allowing models that require millions of time steps (such as RL models) to train quickly. As a result, users can quickly instantiate and train models in just a few lines of code. In addition, users can also directly interface with the game environments via the *get_state()* and *execute_action()* functions, allowing them to train other types of non-RL agents.

Finally, the Python version of the game provides access to metrics tracking, which logs common game metrics such as the number of rounds, the number of deaths, etc. Users can analyze these logs to determine how their agents are progressing during learning or performing during testing. They help facilitate an empirical analysis of learned tasks, and can be especially useful when evaluating

²<https://strong-tact.github.io/game/>

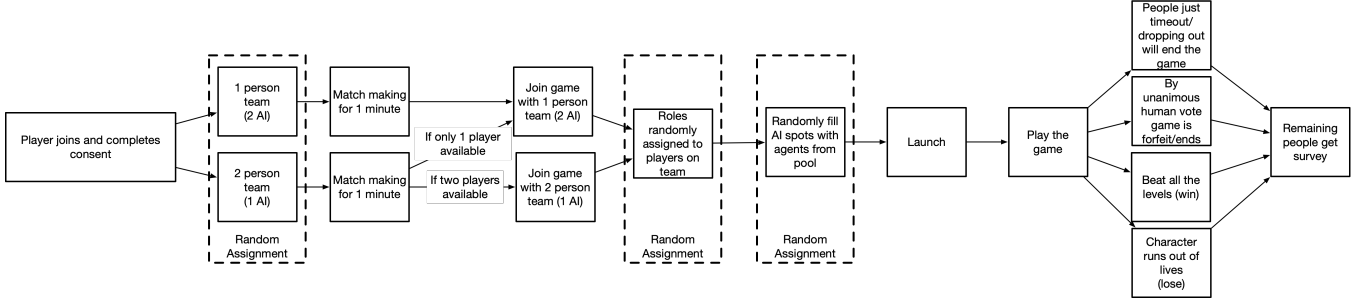


Figure 5: Workflow of the matchmaking system when there are one or two AI players in each game. We are using a separate build for the hybrid-teaming conditions as the game server needs to be connected to an AI interface to draw agents and create instances from before getting launched.

performance against the team scoring function we plan to use during the competition.

4.2 Baseline Agent

To provide participants with reference implementations and enable testing of human-AI teaming scenarios, we developed two baseline agents using different approaches. The first baseline was a reinforcement learning agent trained using the Proximal Policy Optimization (PPO) [24] algorithm, which has shown strong performance in similar partially observable environments. This agent was intentionally trained with a sparse reward signal that only provides positive reinforcement upon the team reaching the tower, forcing it to learn the implicit value of sub-goals like shrine collection through exploration. Training was conducted in the low-fidelity Python environment using self-play, where multiple instances of the agent learned to coordinate with copies of itself.

The second baseline was a Hierarchical Task Network (HTN) [4, 17] agent that incorporates hand-authored domain knowledge about the game’s mechanics and collaborative aspects. The underlying HTN model decomposes the high-level goal of level completion into subtasks, such as reaching the shrine and reaching the tower. Notably, the HTN agent was designed with explicit communication behavior—it can both generate and interpret pin placements. For example, when discovering a monster for the first time, the agent placed a ping at the monster’s location to alert its teammates.

The two baseline agents represented distinct approaches to the challenge of developing AI teammates - one leveraged carefully crafted domain knowledge and explicit communication protocols, the other learned behaviors purely through environmental interaction. Initial testings showed that both agents could successfully participate in mixed human-AI teams, though they displayed different strengths. The HTN agent tended to be more predictable and interpretable to human teammates due to its rule-based nature, whereas the RL agent’s actions sometimes appeared more erratic. Additionally, while the RL agent developed effective pathfinding and combat behaviors, it exhibited notably different communication patterns compared to the HTN agent since it wasn’t explicitly trained to utilize the pinning system. By providing these contrasting baselines, we hoped to inspire participants to explore diverse approaches to agent development while ensuring the competition has a solid foundation for studying human-AI teaming dynamics.

5 Competition as a Pilot Study

5.1 Competition Tracks and Rules

The competition had two tracks: the agent track and the player track. Participants could submit their trained AI agents to the agent track before the submission deadline. The submitted agents needed to be able to play any of the three characters on a team with human players. The goal of the agent track was to raise the awareness of Human-AI teaming [26] by encouraging participants to design and develop online learning agents in addition to traditional offline pre-trained agents. As part of the starter code and packages, we released a low-fidelity version of *Dice Adventure* (introduced in Section 4.1) developed in Python³ and a customized OpenAI Gym environment for participants to train their agents, test the connection to the Unity game environment, and learn to play the game.

For participants who were interested in game play but did not wish to submit agents, we offered a *player track* where they could participate by joining the competition and playing alongside other human and AI players. When players connected, they were assigned to one of three possible conditions: (1) all-human condition; (2) two human one AI condition; or (3) one human two AI condition. We first ran the all-human condition by disabling all AI agents. Then we ran the conditions where AI agents were introduced onto the teams. Characters were randomly assigned to human and AI players in each of the three conditions. In condition (1), the players knew that they were going to play with human players. In conditions (2) and (3) (workflow shown in Figure 5), the players knew that they were to play with AI agents but did not know the number of AI players on the team, nor did they know which characters were controlled by AI agents. Due to technical issues, only the HTN agent was available during the competition. We revised the RL agent to fix the technical issue and made it available after the competition for future studies and iterations of the competition.

During the competition, *Dice Adventure* was available online at [itch.io](https://github.com/STRONG-TACT/Dice-Adventure-Agents).⁴ We also released a Windows build of the game for developers and players. There were two modes of the game, the practice mode and the competition mode. In the practice mode, players could switch between three characters. This made it easier to learn how to play the game and improve their familiarity and skill. The

³<https://github.com/STRONG-TACT/Dice-Adventure-Agents>

⁴<https://cmu-tact.itch.io/dice-adventure>



Figure 6: The consent form page, which introduces the study design, risks, and benefits before entering the competition.

competition mode was only for virtual and in-person matchmaking events.

When participants registered for the competition, they were assigned an anonymized competition ID to use for logging into the game. Non-competitors were also assigned a random ID when first connecting. Upon entering the competition, all players read through a consent form (Figure 6) and then entered their competition ID or random ID (Figure 7) into the text box to participate. We only collected gameplay data during the competition. All logged data was associated with the anonymized identifiers, no personally identifiable information (PII) was collected through the game. After the competition, we reported the winner’s identifiers to the competition organizers and they used these to contact the winners.

At the end of each game, players were asked to fill out a short survey with the following five 5-points Likert scale questions:

- I would play with this team again.
- Our team was efficient.
- Our team communicated well.
- Our team performed well.
- Our team got better over time.

We chose these questions to quickly (< 30 sec.) assess each player’s perception of their team after each game with respect to the team science constructs of communication, coordination, and adaptation [26, 27]. The design of our study was reviewed and approved by our institutional review board.

5.2 Evaluation Criteria

To evaluate human-AI teaming efficiency, submitted agents were put into the agent pool and randomly assigned to teams during the matchmaking events. Participants who signed up for the player track could play as many games as they liked during the competition sessions. We set up a real-time leaderboard on the competition website to display the scores of each team. After the submission portal closed at the conclusion of the last matchmaking event, the top three teams on the leaderboard were awarded cash prizes sponsored by the IEEE Computational Intelligence Society Education Competition Fund. The cash prize for each team was distributed equally among the three team members, with the developers of winning AI agents receiving the prize.

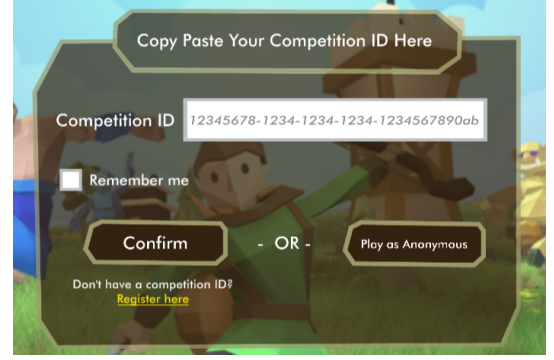


Figure 7: The screen where players entered their Competition IDs, which were required if a player was interested in getting on the leaderboard and competing for the prizes. Players could also play as anonymous.

To quantitatively compare the performance of different teams and determine the winners, we proposed a scoring function to calculate each team’s score. Equation 1 represents the score S that a team earned at each level i they cleared. Since *Dice Adventure* is a turn-based game, the number of rounds a team took to clear a level reflected the team’s efficiency. Therefore, we set the initial score to 100 at the start of each level and applied a decaying factor d ($d=0.95$) to model the exponential decay of the score as players took r rounds to complete a level.

In addition to the base score, there is a reward component for each character. Each character has three lives, with each life consisting of three health points. A character loses a health point when an enemy defeats them or when they take damage from the environment. When a character loses all three health points, they lose a life or “die”, but will respawn after waiting for a whole round. The characters’ health is refreshed and reset to 3 at the start of each level, but their lives carry on over the levels. At the end of each level, bonus points are awarded to the characters ($j=1, 2, 3$) that did not die ($rs=1$). These bonus points, with a maximum of 10, are calculated proportionally based on the remaining health h of the character j , divided by their default (maximum) health h_{max} at the end of a level i . Lastly, Equation 2 is the sum of scores the team achieved throughout the number of levels n they survived.

$$S_i = 100 \times d^{r_i} + \sum_{j=1,2,3} \frac{h_{ij}}{h_{max_{ij}}} \times rs_{ij} \times 10 \quad (1)$$

$$\text{team_score} = \sum_{i=1}^n S_i \quad (2)$$

6 Discussion

6.1 Competition Summary

We hosted a two-day in-person session for the competition, followed by online matchmaking events lasting twelve days. The competition was advertised through multiple sources, such as on-campus ads, mailing lists and social platforms. A total of 21 people signed up to play the game during the in-person session, 12 of

Metric	Hybrid	All-human	All teams
# Teams	38	18	56
# Avg games played	12.97	19.67	15.13
# Avg cleared levels	1.92	2.94	2.25
Avg total score	138.91	215.79	163.62
Highest score	322.36	405.90	405.90
Lowest score	44.01	85.74	44.01

Table 1: Summary statistics on team scores.

Level name	# Teams	Avg rounds	Avg score
level1	8	1.5	92.6546875
level2	6	3.16	85.06
Medium_2	6	6.50	71.86
Large_1	5	7.00	71.15
HackathonLarge1	6	9.33	63.72
HackathonLarge2	5	14.00	49.08

Table 2: Summary statistics on number of teams, average rounds taken and average score for each level that the all-human teams have completed.

whom competed for the prizes. Several players participated anonymously; they were included in the score calculations to determine the winning teams, but were excluded from the prize competition.

We received two agent submissions after the in-person session. One participant submitted a Monte Carlo tree search (MCTS) agent and the other participant submitted a reinforcement learning (RL) agent. Both agents were integrated into our system and were capable of playing any of the three characters in the game. Unfortunately, no participants joined the online matchmaking sessions, so the two submitted agents were not evaluated with human players. The 2024 competition report, along with the winner announcement, is available on the competition website.⁵

6.2 Game Analysis

During the in-person session, 21 human players and the HTN baseline agent formed 56 unique teams and played 75 games. Of these, 38 hybrid teams (2 AI agents and 1 human player) played 56 games and 18 all-human teams played 19 games. Table 1 presents the summary statistics of the competition. The results show that all-human teams played more games, cleared more levels and achieved a higher total score comparing to the hybrid teams.

Table 2 and Table 3 demonstrate the number of teams that cleared each level, the average number of rounds taken, and the average scores earned by the all-human teams and hybrid teams at the end of each level. Slightly different levels were chosen for all-human teams and hybrid teams to account for practice effects. All the levels were randomly generated using a set of parameters, with levels at the same order exhibiting similar difficulty. The levels are ordered by increasing difficulty.

Level name	# Teams	Avg rounds	Avg score
level1	27	1.93	90.71
level2	18	4.22	81.18
interdependence	11	16.09	53.12
Video	6	11.83	57.12
DistantSpawns	2	23.5	32.2
Jailbreak	0	0	0

Table 3: Summary statistics on number of teams, average rounds taken and average score for each level that the hybrid teams have completed.

All-human			Hybrid	
# Completed	# Teams	Proportion	# Teams	Proportion
6	5	0.28	0	0
5	0	0	2	0.05
4	1	0.06	2	0.05
3	1	0.06	6	0.16
2	5	0.28	8	0.21
1	6	0.33	20	0.53

Table 4: Summary statistics on the number of levels that all-human and hybrid teams completed in a game.

Table 4 presents the level completion rates in all-human teams and hybrid teams, highlighting the number of teams that successfully completed each level. By conducting the Chi-square test, we found a statistically significant difference ($p < 0.05$) in the distribution of the number of completed levels between all-human teams and hybrid teams. However, we only found a significant difference ($p < 0.01$) between the proportion of two types of teams that completed 6 levels. The results suggest that more statistical power is needed for future experiments.

We aim to conduct further comprehensive analysis on how teams used pings to communicate, approached challenges, and rated their teaming experiences at the end of each game.

6.3 Game Observations

We observed how different teams played the game differently during the competition. In all-human teams, we noticed that team members usually had similar interpretations of the pings, making it more efficient for them to convey information to their teammates. For example, some players flagged their teammates' shrines (personal goals) when such objects were in their view but outside of the range of their teammates' views. We also observed that players on all-human teams tended to employed different strategies at various stages of a game. Specifically, in the first few rounds of a level, some players focused on exploring the map and locating their personal goals. Once a player found their personal goal, their behaviors varied depending on their character's abilities. For instance, some players who controlled the *giant* challenged the enemies to clear paths for their teammates. On the other hand, some other players who controlled the *dwarf* continued exploring the map and used pings to communicate with their teammates.

⁵https://strong-tact.github.io/2024_competition/

An interesting observation from hybrid teams was that human players placed fewer pings after realizing that the HTN agent struggled to “understand” the meaning and respond as expected. This insight led us to consider the feasibility of training agents equipped with theory-of-mind capabilities [30]. We believe that human players would benefit more if mutual understanding can be established between them and the AI agents. Furthermore, such mutual understanding could also improve team synchronization in using pings by enabling AI agents to imitate human behaviors.

7 Future Extensions

7.1 Proposed Experimental Design

The *Dice Adventure* Competition was designed as a pilot study to investigate human-AI teaming effects when role assignments, team compositions, and team dynamics change. Additionally, we aimed to explore whether teams perform better with different types of models (pre-trained models versus online-learning models).

The data collected from the all-human session served as a “control” condition for comparison with results from hybrid conditions. In addition, we planned to conduct all-agent condition experiments after the competition, as it does not require human participation. We also aimed to recruit participants from sourcing platforms to conduct randomized controlled trials, controlling for factors such as the number of AI agents on a team, agent types, and the characters the agents play.

7.2 Prospective Analyses

The asymmetric design of *Dice Adventure* provided a unique opportunity to investigate how different team compositions can affect collaboration outcomes. We hypothesized that human-AI teams would outperform both pure AI or pure human teams due to complementary strengths. AI agents, with their substantial computational power, can efficiently navigate through complex state spaces in microseconds, while humans bring adaptability and strategic thinking to handle unforeseen challenges. Moreover, with this asymmetry in abilities, we are particularly interested in analyzing how the same AI agents perform when assigned different roles in the game. Similar to the pilot study, the team’s score will be used as an indicator to evaluate and compare different teams’ performance.

By analyzing both quantitative and qualitative data as well as conducting case studies on individual games, we aim to develop an evaluation framework for human-machine collaboration. This framework will address questions such as what defines successful human-machine teaming from the human perspective, whether team competitiveness directly leads to positive agent ratings, and crucially, how we can computationally and automatically identify collaborative behaviors in real-time for future online agent training.

We aim to run multiple iterations of the *Dice Adventure* Competition through different AI and games conferences and institutions to collect agent submissions and game play data. By conducting comprehensive analyses of game play, we want to understand the strategies employed by humans and agents on winning teams, and examine how these strategies can be characterized and reproduced to enable more effective human-AI teaming on this and other tasks. We also plan to compare the strategies we identify with those identified in prior human-AI teaming work [9]. We will also run offline

evaluations of all-AI teams and compare the teaming performance of different mixtures of agents. We hope to reveal and bring valuable insights into how AI agent can be designed to better support teaming; e.g., we hypothesize that agents that adapting their behaviors to their teammates on the fly will perform better than those that employ a fixed set of pre-defined behaviors.

8 Contributions, Challenges and Opportunities

In this work, we developed a game that can be used as a testbed to investigate human-AI teaming. Unlike prior games, such as *Overcooked*, our game features information asymmetry, ability asymmetry, and role asymmetry, which makes it uniquely well-suited for studying aspects of teaming, including role assignment, team composition, communication, coordination, and adaptation. We also provided a low-fidelity Python version of the game to support AI training. Our goal for developing *Dice Adventure* and hosting the competition is not limited to providing an environment for participants to build agents that can play the game on their own. We would like to invite the participants and general audience to think about approaches to collaborate with AI agents on complex asymmetric tasks. Therefore, we evaluated human-AI teaming effects that are not discussed in existing studies. In addition to elevating the awareness of human-AI teaming and encouraging participants to explore teaming dynamics, we wanted to use this competition as an opportunity to pilot an experiment on human-AI teaming and further refine our research program to better understanding human-AI teaming.

As it was the first time for us to host the *Dice Adventure* competition and to run the human-subject studies, we did not have a reasonable expectation on the types or volume of agents submitted. As a result, we only explored open-ended questions about teaming strategies that different human-AI teams employed. Such limitations lead us to the challenge of generalizing the findings concluded from a game-specific environment to more complex real-life scenarios with less environmental control. Nevertheless, we believe that this work will contribute to the broader scholarship on AI, Human-Computer Interaction, and Human-AI-Interaction. It also provides new paradigms for game development, experimental design, and competition planning.

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